1.1 Methodology

AIM & OBJECTIVE:
To Forecast the Demand and to determine optimal Procurement plan

PROCEDURE
Step 1: To Perform ABC analysis of last 12 months and classify them, In A (Always), B (Better) and C (Control) categories based on their annual usage value.

Step 2: To carry out forecasting by taking their last three years monthly consumption as base, for some items belonging to categories of ABC analysis

Step 3: Apply the optimal procurement policy for Forecasted demands.

Step 4: Results and recommendations.

STEP-BY-STEP PROCECURE USED FOR SOLVING THE PROBLEM

- First of all, we collect data related to sales of all types of Tiles and Stones sold by the company.
- Then we plot the data to study the behavior.
- Next we Study the plots for if it is seasonal or non-seasonal.
- Then we apply the appropriate Forecasting method for various months of the next year.
- If the methods we are using do not give good Forecasts, we calculate the error and apply a different method until the good Forecast is obtained.
- Here we have used Time Series Forecasting Methods to Forecast the demand.
- Collect data for Inventory carrying cost per annum and Ordering cost per order.
- Fit the appropriate Procurement Plan.
- Looking into the results carefully conclude and recommend plan for the procurement of the various types of Tiles.

1.2 ABC ANALYSIS

- Procedure for ABC Analysis

The steps for classification of products into A, B and C categories are as follows:

Step 1: Determine the number of units sold for each product in the past

one year.

Step 2: Determine the unit cost for each product.

Step 3: Compute the annual usage value of each product by multiplying the units sold and unit cost price.

Annual usage value = Total units sold * unit cost of product

Step 4: Sort the items in descending order by usage value as computed in previous step.

Step 5: Calculate the cumulative sum of the no. of items and the usage value for each item obtained in Step 3.

Step 6: Calculate the percentage value obtained in step 5 with respect to grand total of the corresponding columns.

Step 7: Draw a line graph having cumulative percent of items on x-axis and cumulative percent of annual consumption on y-axis.

Step 8: Mark the points on graph where the slope of the curve changes sharply. Such points are referred as points of inflexion.

Step 9: Lastly, the usage value and percent of items corresponding to these points determine the classification of items as A, B or C.

List of Items

S No.	Item Code	Quantity (in boxes)
1	LSS650478	15624
2	AAEB329176	**13652**
3	ACEB405938	14658
4	ASP763504	12354
5	AAE914763	10856
6	AC693208	9958
7	WS159847	9875
8	MS653209	9954
9	CCMSCO527104	14625
10	LSS290453	9562
11	RS849672	8698
12	RS831407	5545
13	TPS803214	5126
14	RS503821	4521
15	TT384920	2895
16	MS145287	2954
17	SS453209	1874
18	AFROM712345	1574
19	XS701593	1743
20	MLS502894	2109
21	LS958701	1276
22	CLLB874509	1743
23	RPS357109	1426
24	ABHT394725	1584
25	LPSC726341	1289
26	TRS274839	2071
27	CMP309182	1989
28	WS590634	1475
29	PS689037	1954
30	DAM736592	1352
31	DF197632	2275
32	LRS805432	1922
33	SCTWL746302	1819
34	APS354809	1846
35	AD819206	2381
36	AS296813	2541
37	AS918206	2421
38	SS146328	1174
39	SSM130289	2346
40	CMV302157	1883
41	CS468250	1757
42	AG907346	1731
43	AF872954	1541
44	ADEB590147	1923
45	CS680412	1727
46	LS508493	1934

47	BS274608	1906
48	AS620589	1906
49	CCMSMS476302	1450
50	DCM905817	954
51	LS407592	1007
52	ASRM928150	1428
53	AA320589	1255
54	MM927513	1580
55	SMP983215	989
56	LSS940685	921
57	SCP642917	1541
58	MASP975104	1250
59	AB738205	1200
60	LP507831	1181
61	MMS307914	1229
62	GS258931	1364
63	RS746301	1440
64	ACISB917634	1102
65	H731604	856
66	TPS874502	1205
67	PS158490	1371
68	STCS621890	1181
69	SS109284	1277
70	APS609438	1161
71	NPS685102	1210
72	CRM841236	831
73	ASB569083	1071
74	BMP746238	1101
75	MMS248309	662
76	AD738210	1386
77	HP913205	1339
78	CMPP364820	866
79	HS495307	894
80	ISMC328615	1152
81	ADM429715	877
82	SS915870	938
83	RPS872503	781
84	AF236791	981
85	AC731028	1023
86	ACD873215	854
87	PGS407218	978
88	CBS320761	885
89	DAMB365802	679
90	ADM591827	746
91	SS629104	536
92	AGL579318	695
93	AS512609	651
94	RCT869150	548
95	GS401295	569

CALCULATIONS

S No.	Item Code	Demand	Unit Cost	Annual Usage	Usage (in %)	Cumulative Item (in%)	Cumulative Value(in%)	Category
1	LSS650478	15624	1025.32	16019600	7.62%	1.052632	7.62%	A
2	AAEB329176	13652	1437.25	19621337	9.97%	2.105263	17.59%	A
3	ACEB405938	14658	1154.326	16920111	11.44%	3.157895	29.03%	A
4	ASP763504	12354	989.5641	12225075	8.68%	4.210526	37.71%	A
5	AAE914763	10856	965.3254	10479573	6.40%	5.263158	44.11%	A
6	AC693208	9958	875.6521	8719744	5.90%	6.315789	50.01%	A
7	WS159847	9875	964.32	9522660	6.15%	7.368421	56.17%	A
8	MS653209	9954	962.3258	9578991	3.98%	8.421053	60.14%	A
9	CCMSCO527104	8625	1325.127	11429224	1.80%	9.473684	61.94%	B
10	LSS290453	9562	1425.365	13629344	1.39%	10.52632	63.33%	B
11	RS849672	8698	785.2699	6830278	1.37%	11.57895	64.70%	B
12	RS831407	5545	782.6254	4339658	1.31%	12.63158	66.01%	B
13	TPS803214	5126	695.46	3564928	1.20%	13.68421	67.21%	B
14	RS503821	4521	545.8448	2467764	1.16%	14.73684	68.37%	B
15	TT384920	2895	805.6699	2332414	1.10%	15.78947	69.47%	B
16	MS145287	2954	575.2949	1699421	0.80%	16.84211	70.28%	B
17	SS453209	1874	745.527	1397118	0.66%	17.89474	70.93%	B
18	AFROM712345	1574	875.6812	1378322	0.65%	18.94737	71.58%	B
19	XS701593	1743	785.2107	1368622	0.65%	20	72.23%	B
20	MLS502894	2109	635.3863	1340030	0.63%	21.05263	72.86%	B
21	LS958701	1276	1005.546	1283077	0.61%	22.10526	73.47%	B
22	CLLB874509	1743	705.1137	1229013	0.58%	23.15789	74.05%	B
23	RPS357109	1426	855.6592	1220170	0.58%	24.21053	74.62%	B
24	ABHT394725	1584	765.4657	1212498	0.57%	25.26316	75.20%	B
25	LPSC726341	1289	925.4806	1192944	0.56%	26.31579	75.76%	B
26	TRS274839	2071	575.6665	1192205	0.56%	27.36842	76.32%	B
27	CMP309182	1989	585.9229	1165401	0.55%	28.42105	76.87%	B
28	WS590634	1475	785.3816	1158438	0.55%	29.47368	77.42%	B
29	PS689037	1954	585.61	1144282	0.54%	30.52632	77.96%	B
30	DAM736592	1352	825.5712	1116172	0.53%	31.57895	78.48%	B
31	DF197632	2275	485.6869	1104938	0.52%	32.63158	79.01%	B
32	LRS805432	1922	565.5344	1086957	0.51%	33.68421	79.52%	B
33	SCTWL746302	1819	585.2458	1064562	0.50%	34.73684	80.02%	B
34	APS354809	1846	575.87	1063056	0.50%	35.78947	80.52%	B
35	AD819206	2381	445.2098	1060045	0.50%	36.84211	81.02%	B
36	AS296813	2541	415.1558	1054911	0.50%	37.89474	81.52%	B
37	AS918206	2421	435.7241	1054888	0.50%	38.94737	82.02%	B
38	SS146328	1174	895.2721	1051049	0.50%	40	82.51%	B
39	SSM130289	2346	445.4275	1044973	0.49%	41.05263	83.01%	B
40	CMV302157	1883	545.8507	1027837	0.48%	42.10526	83.49%	B
41	CS468250	1757	575.2494	1010713	0.48%	43.15789	83.97%	B
42	AG907346	1731	575.7722	996661.7	0.47%	44.21053	84.44%	B
43	AF872954	1541	625.2928	963576.2	0.45%	45.26316	84.89%	B
44	ADEB590147	1923	495.2182	952304.6	0.45%	46.31579	85.34%	B
45	CS680412	1727	545.2047	941568.5	0.44%	47.36842	85.79%	B
46	LS508493	1934	485.91	939749.9	0.44%	48.42105	86.23%	B

47	BS274608	1906	485.7991	925933.1	0.44%	49.47368	86.67%	B
48	AS620589	1906	485.6135	925579.3	0.44%	50.52632	87.11%	B
49	CCMSMS476302	1450	625.8171	907434.8	0.43%	51.57895	87.53%	B
50	DCM905817	954	925.2398	882678.8	0.42%	52.63158	87.95%	B
51	LS407592	1007	825.3672	831144.8	0.39%	53.68421	88.34%	B
52	ASRM928150	1428	575.192	821374.2	0.39%	54.73684	88.73%	B
53	AA320589	1255	625.6493	785189.9	0.37%	55.78947	89.10%	B
54	MM927513	1580	495.8953	783514.6	0.37%	56.84211	89.47%	B
55	SMP983215	989	785.9649	777319.3	0.37%	57.89474	89.84%	B
56	LSS940685	921	825.2574	760062.1	0.36%	58.94737	90.20%	B
57	SCP642917	1541	485.5778	748275.4	0.35%	60	90.55%	B
58	MASP975104	1250	575.8556	719819.5	0.34%	61.05263	90.89%	B
59	AB738205	1200	585.5321	702638.5	0.33%	62.10526	91.22%	B
60	LP507831	1181	575.5453	679719	0.32%	63.15789	91.54%	B
61	MMS307914	1229	545.6162	670562.3	0.32%	64.21053	91.86%	B
62	GS258931	1364	485.2821	661924.8	0.31%	65.26316	92.17%	B
63	RS746301	1440	455.7563	656289.1	0.31%	66.31579	92.48%	B
64	ACISB917634	1102	585.3111	645012.8	0.30%	67.36842	92.78%	B
65	H731604	856	745.8619	638457.8	0.30%	68.42105	93.08%	B
66	TPS874502	1205	525.8236	633617.4	0.30%	69.47368	93.38%	B
67	PS158490	1371	455.3893	624338.7	0.29%	70.52632	93.68%	B
68	STCS621890	1181	525.9025	621090.9	0.29%	71.57895	93.97%	B
69	SS109284	1277	485.3638	619809.6	0.29%	72.63158	94.26%	B
70	APS609438	1161	525.8883	610556.3	0.29%	73.68421	94.55%	B
71	NPS685102	1210	485.1365	587015.2	0.28%	74.73684	94.83%	B
72	CRM841236	831	705.4802	586254	0.28%	75.78947	95.11%	C
73	ASB569083	1071	545.1048	583807.2	0.28%	76.84211	95.38%	C
74	BMP746238	1101	525.6042	578690.2	0.27%	77.89474	95.65%	C
75	MMS248309	662	855.7277	566491.7	0.27%	78.94737	95.92%	C
76	AD738210	1386	405.4561	561962.2	0.27%	80	96.19%	C
77	HP913205	1339	405.8575	543443.2	0.26%	81.05263	96.44%	C
78	CMPP364820	866	625.921	542047.6	0.26%	82.10526	96.70%	C
79	HS495307	894	575.917	514869.8	0.24%	83.15789	96.94%	C
80	ISMC328615	1152	445.1889	512857.6	0.24%	84.21053	97.18%	C
81	ADM429715	877	575.314	504550.4	0.24%	85.26316	97.42%	C
82	SS915870	938	495.8333	465091.6	0.22%	86.31579	97.64%	C
83	RPS872503	781	585.1471	456999.9	0.22%	87.36842	97.86%	C
84	AF236791	981	465.2087	456369.7	0.22%	88.42105	98.07%	C
85	AC731028	1023	445.79	456043.2	0.22%	89.47368	98.29%	C
86	ACD873215	854	525.6899	448939.2	0.21%	90.52632	98.50%	C
87	PGS407218	978	445.539	435737.1	0.21%	91.57895	98.70%	C
88	CBS320761	885	485.4828	429652.3	0.20%	92.63158	98.91%	C
89	DAMB365802	679	625.8483	424951	0.20%	93.68421	99.11%	C
90	ADM591827	746	545.1669	406694.5	0.19%	94.73684	99.30%	C
91	SS629104	536	745.3107	399486.5	0.19%	95.78947	99.49%	C
92	AGL579318	695	415.3734	288684.5	0.14%	96.84211	99.62%	C
93	AS512609	651	415.5736	270538.4	0.13%	97.89474	99.75%	C
94	RCT869150	548	485.357	265975.6	0.13%	98.94737	99.88%	C
95	GS401295	569	455.3074	259069.9	0.12%	100	100.00%	C

RESULTS OF ABC ANALYSIS

We will be selecting the 8 Products from the A category for the purpose of forecasting and procurement. The calculation and the corresponding chart for the same are shown below. The number of items in each category are as follows:

- Class A: 8 Items
- Class B: 63 Items
- Class C: 24 Items

These selected class A items are listed as follows:

S No.	Item Code
1	LSS650478
2	AAEB329176
3	ACEB405938
4	ASP763504
5	AAE914763
6	AC693208
7	WS159847
8	MS653209

1.3 Demand Forecasting

1.3.1 Product 1:

LSS650478

Using the sales data of past three years for the product LSS650478, we'll prepare a time series forecasting model based on the features of data.

Sales Data Table

Months	2022-23	2023-24	2024-25
Apr	749	1329	1360
May	862	1355	1450
Jun	532	1080	1180
Jul	712	1363	1502
Aug	1366	1528	1562
Sep	1515	1606	1573
Oct	862	1001	1081
Nov	1104	1070	1092
Dec	857	957	896
Jan	1218	1100	1238
Feb	1280	1275	1257
Mar	1332	1290	1274

Following is the stepwise approach that we will be following to solve our problem.

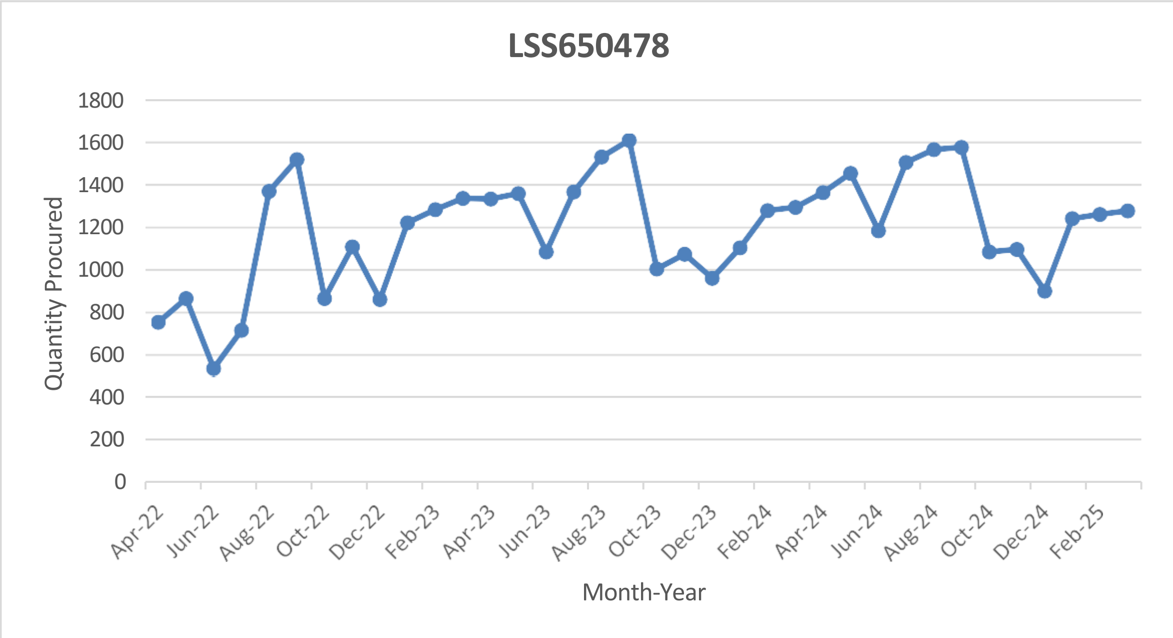
1. **Data Analysis**: The main aim of this step is to identify level, trend, seasonality and irregular behavior and also to identify if the time series is stationary or not. For this we'll plot data into different information charts to extract meaningful insights.

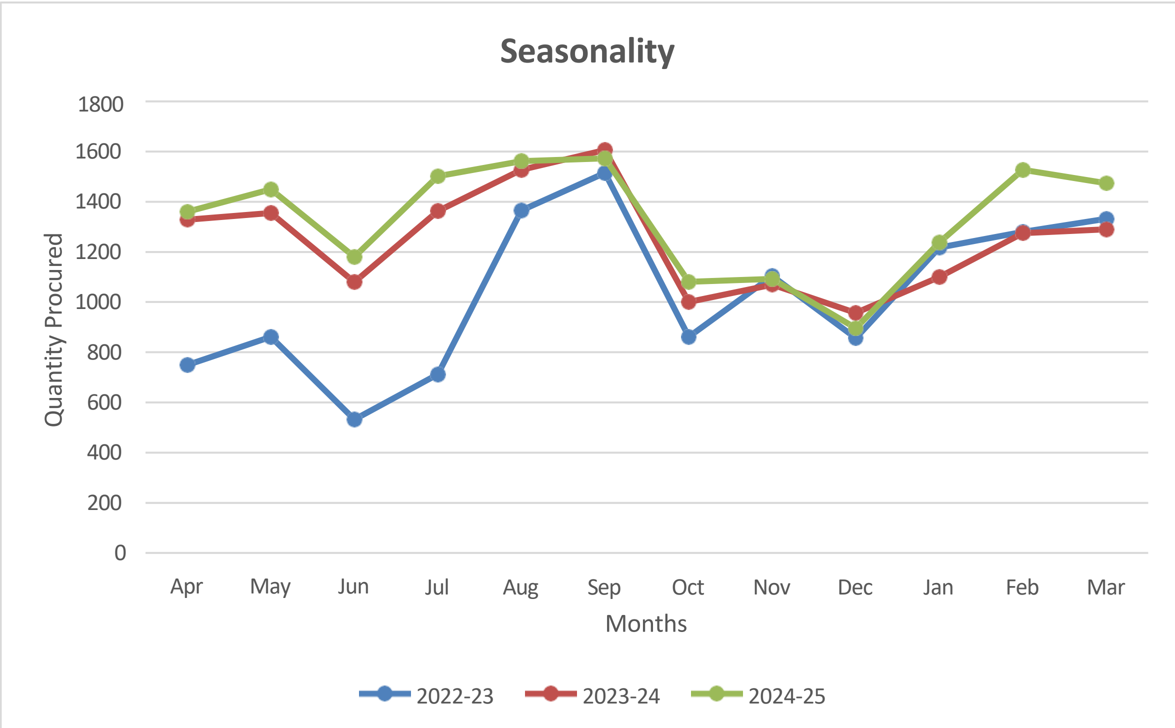
2. **Fitting Model**: Now we'll fit various statistical models on the data collected. The main aim of this step is to fit a model in order to forecast the future demand. To compare actual V/s predicted values we plot a line graph.

3. **Selecting Best Model**: Best model on the basis of 'MAPE' metric is used to compare the fitted models. Model having lowest MAPE is considered as best fitted model.

4. **Forecasting Demand**: The final step in process to forecast the demand for next period. We will forecast the demand data for upcoming year that will serve as a benchmark for the firm for their purchase orders next year.

i) Data Analysis





An initial examination of the Demand data reveals the presence of a **Trend component**, a **Seasonal component**, and some degree of **irregular variation**. This indicates that the demand series is **non-stationary**. As such, forecasting models capable of handling trend and seasonality are required.
To identify the most suitable forecasting method, we have selected two models for comparison:

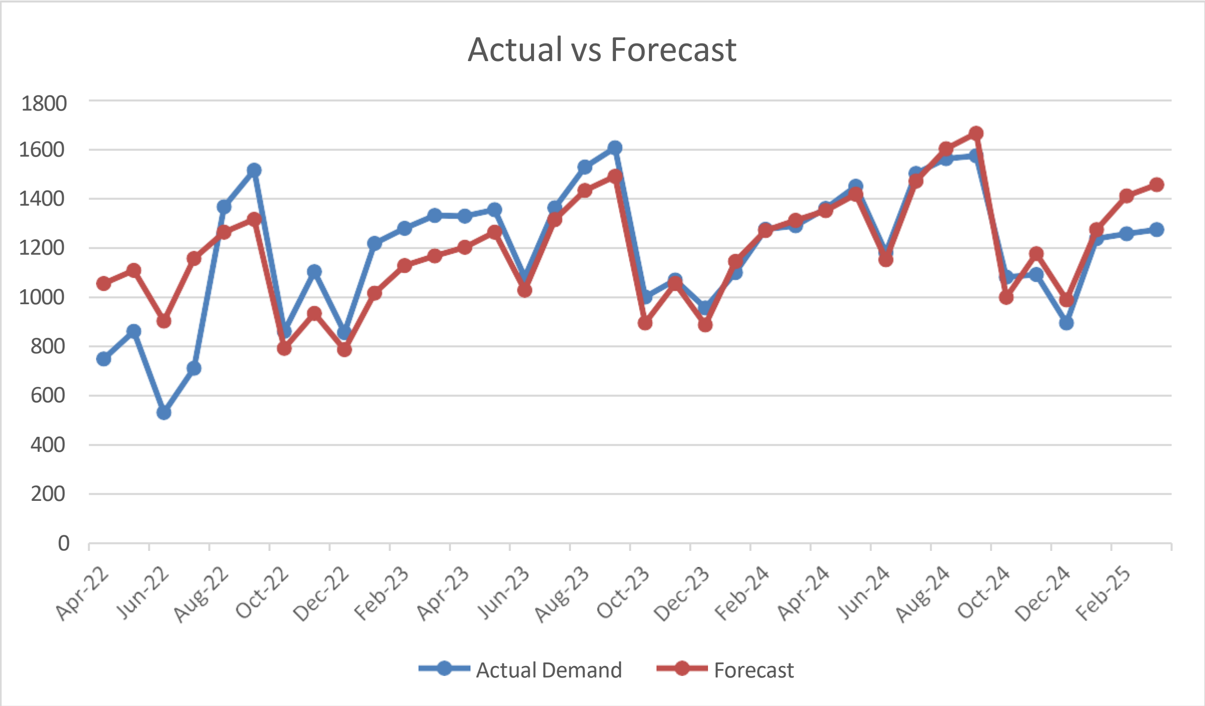
- **Decomposition Method**
- **Holt-Winters Exponential Smoothing**

Both models will be applied to the complete demand dataset. Their performance will be evaluated using the **Mean Absolute Percentage Error (MAPE)** metric. The model that yields the **lowest MAPE** will be considered the most accurate and will be used for final demand forecasting.

(ii) Fitting Model

Model 1 : Decomposition Method

Time Period	Month-Year	Demand	12 Point Moving Average	Center Moving Average	Ratio to Moving Average	Adjusted Seasonal Indices	Deseasonalize	Trend	Forecast	Error %
1	Apr-22	749				106.49	703.38	990.98	1055.25	40.89%
2	May-22	862				110.67	778.88	1002.55	1109.54	28.72%
3	Jun-22	532				89.12	596.94	1014.13	903.80	69.89%
4	Jul-22	712				112.86	630.87	1025.70	1157.61	62.59%
5	Aug-22	1366				121.79	1121.58	1037.28	1263.33	7.52%
6	Sep-22	1515	1032.42			125.51	1207.06	1048.86	1316.44	13.11%
7	Oct-22	862	1080.75	1056.58	0.82	74.70	1153.92	1060.43	792.16	8.10%
8	Nov-22	1104	1121.83	1101.29	1.00	87.18	1266.31	1072.01	934.60	15.34%
9	Dec-22	857	1167.50	1144.67	0.75	72.61	1180.25	1083.58	786.81	8.19%
10	Jan-23	1218	1221.75	1194.63	1.02	92.76	1313.03	1095.16	1015.90	16.59%
11	Feb-23	1280	1235.25	1228.50	1.04	101.94	1255.65	1106.74	1128.20	11.86%
12	Mar-23	1332	1242.83	1239.04	1.08	104.36	1276.36	1118.31	1167.06	12.38%
13	Apr-23	1329	1254.42	1248.63	1.06	106.49	1248.05	1129.89	1203.17	9.47%
14	May-23	1355	1251.58	1253.00	1.08	110.67	1224.34	1141.46	1263.28	6.77%
15	Jun-23	1080	1259.92	1255.75	0.86	89.12	1211.84	1153.04	1027.60	4.85%
16	Jul-23	1363	1250.08	1255.00	1.09	112.86	1207.69	1164.62	1314.38	3.57%
17	Aug-23	1528	1249.67	1249.88	1.22	121.79	1254.59	1176.19	1432.52	6.25%
18	Sep-23	1606	1246.17	1247.92	1.29	125.51	1279.56	1187.77	1490.79	7.17%
19	Oct-23	1001	1248.75	1247.46	0.80	74.70	1340.00	1199.34	895.93	10.50%
20	Nov-23	1070	1256.67	1252.71	0.85	87.18	1227.31	1210.92	1055.71	1.34%
21	Dec-23	957	1265.00	1260.83	0.76	72.61	1317.97	1222.50	887.68	7.24%
22	Jan-24	1100	1276.58	1270.79	0.87	92.76	1185.82	1234.07	1144.76	4.07%
23	Feb-24	1275	1279.42	1278.00	1.00	101.94	1250.75	1245.65	1269.80	0.41%
24	Mar-24	1290	1276.67	1278.04	1.01	104.36	1236.12	1257.22	1312.02	1.71%
25	Apr-24	1360	1283.33	1280.00	1.06	106.49	1277.16	1268.80	1351.10	0.65%
26	May-24	1450	1285.17	1284.25	1.13	110.67	1310.18	1280.38	1417.01	2.27%
27	Jun-24	1180	1280.08	1282.63	0.92	89.12	1324.04	1291.95	1151.40	2.42%
28	Jul-24	1502	1291.58	1285.83	1.17	112.86	1330.86	1303.53	1471.16	2.05%
29	Aug-24	1562	1290.08	1290.83	1.21	121.79	1282.51	1315.10	1601.70	2.54%
30	Sep-24	1573	1288.75	1289.42	1.22	125.51	1253.27	1326.68	1665.14	5.86%
31	Oct-24	1081				74.70	1447.09	1338.26	999.70	7.52%
32	Nov-24	1092				87.18	1252.55	1349.83	1176.82	7.77%
33	Dec-24	896				72.61	1233.96	1361.41	988.54	10.33%
34	Jan-25	1238				92.76	1334.59	1372.98	1273.62	2.88%
35	Feb-25	1257				101.94	1233.09	1384.56	1411.41	12.28%
36	Mar-25	1274				104.36	1220.79	1396.14	1456.99	14.36%



Model 2 : Holt Winter's Additive Method

Model Description

			Model Type
Model ID	VAR00001	Model_1	Winters' Additive

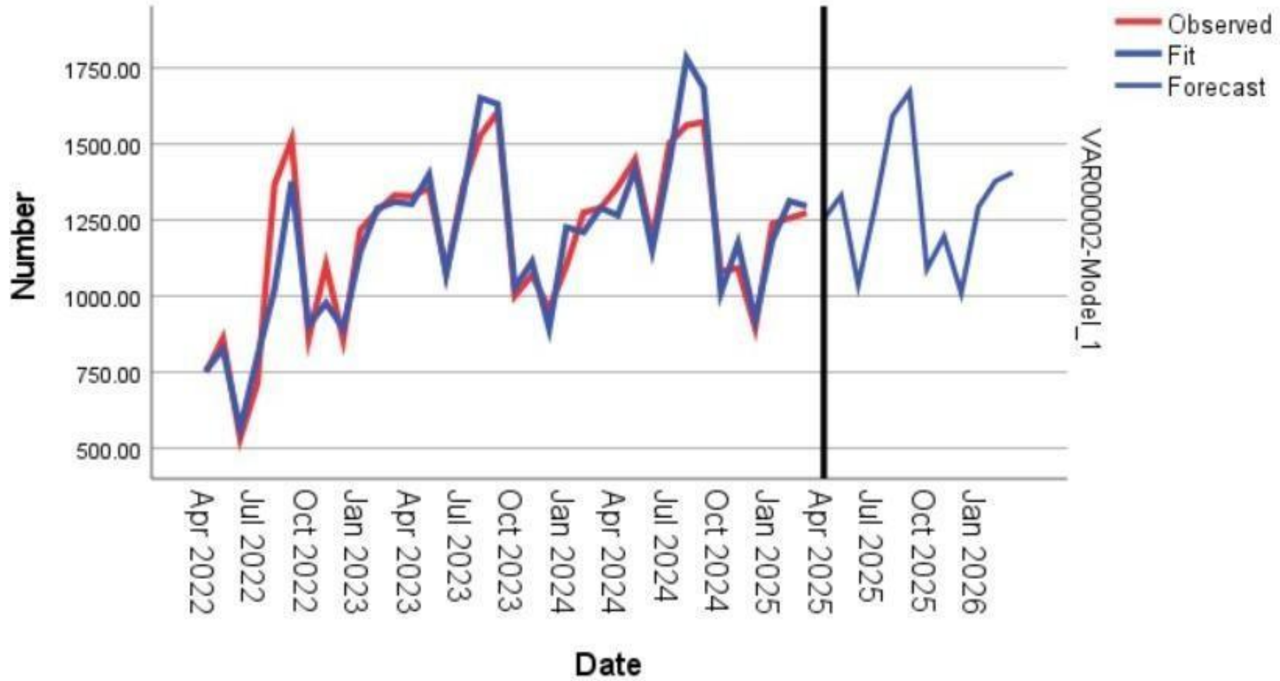
Model Summary

Model Fit

Fit Statistic	Mean	SE	Minimum	Maximum	Percentile 5	10	25	50	75	90	95
Stationary R-squared	.705	.	.705	.705	.705	.705	.705	.705	.705	.705	.705
R-squared	.873	.	.873	.873	.873	.873	.873	.873	.873	.873	.873
RMSE	97.362	.	97.362	97.362	97.362	97.362	97.362	97.362	97.362	97.362	97.362
MAPE	5.509	.	5.509	5.509	5.509	5.509	5.509	5.509	5.509	5.509	5.509
MaxAPE	25.165	.	25.165	25.165	25.165	25.165	25.165	25.165	25.165	25.165	25.165
MAE	66.395	.	66.395	66.395	66.395	66.395	66.395	66.395	66.395	66.395	66.395
MaxAE	343.750	.	343.750	343.750	343.750	343.750	343.750	343.750	343.750	343.750	343.750
Normalized BIC	9.455	.	9.455	9.455	9.455	9.455	9.455	9.455	9.455	9.455	9.455

Model Statistics

Model	Number of Predictors	Model Fit statistics: Stationary R-squared	R-squared	MAPE	Ljung-Box Q(18): Statistics	DF	Sig.	Number of Outliers
VAR00001-Model_1	0	.705	.873	5.509	13.353	15	.575	0



iii. Selecting Best Model

Method	MAPE
Decomposition Method	11.929%
Holt Winter's Additive Exponential Smoothing Method	5.509%

As the Holt-Winters Additive Exponential Smoothing method produced the lowest Mean Absolute Percentage Error (MAPE) among the evaluated models, it has been selected as the most suitable approach for forecasting demand for the upcoming year.

iv. Forecasting Demand

Forecast

Model		Apr 2025	May 2025	Jun 2025	Jul 2025	Aug 2025	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mar 2026
VAR00001-Model_1	Forecast	1253.81	1330.14	1038.47	1300.14	1593.14	1672.47	1089.14	1196.47	1011.14	1293.14	1378.47	1406.47
	UCL	1451.89	1584.25	1338.33	1639.62	1968.09	2079.81	1526.47	1661.88	1503.01	1810.13	1919.41	1970.34
	LCL	1055.72	1076.02	738.62	960.65	1218.19	1265.13	651.80	731.07	519.27	776.15	837.54	842.60

For each model, forecasts start after the last non-missing in the range of the requested estimation period, and end at the last period for which non-missing values of all the predictors are available or at the end date of the requested forecast period, whichever is earlier.

The forecasted demand values for the next year are:

Month	Forecast
Apr-25	1254
May-25	1330
Jun-25	1038
Jul-25	1300
Aug-25	1593
Sep-25	1672
Oct-25	1089
Nov-25	1196
Dec-25	1011
Jan-26	1293
Feb-26	1387
Mar-26	1406

1.3.2 Product 2:

AAEB329176

Using the sales data of past three years for product AAEB329176, we'll prepare a time series forecasting model based on the features of data.

Sales Data Table			
Month	2022-23	2023-24	2024-25
Apr	657	1131	1137
May	687	1175	1192
Jun	752	985	1129
Jul	712	1549	1547
Aug	948	1583	1629
Sep	1273	1626	1671
Oct	987	932	1102
Nov	1126	1045	1131
Dec	1326	1487	1526
Jan	1252	1475	1373
Feb	1205	1239	1219
Mar	1230	1264	1352

Following is the stepwise approach that we will be following to solve our problem.

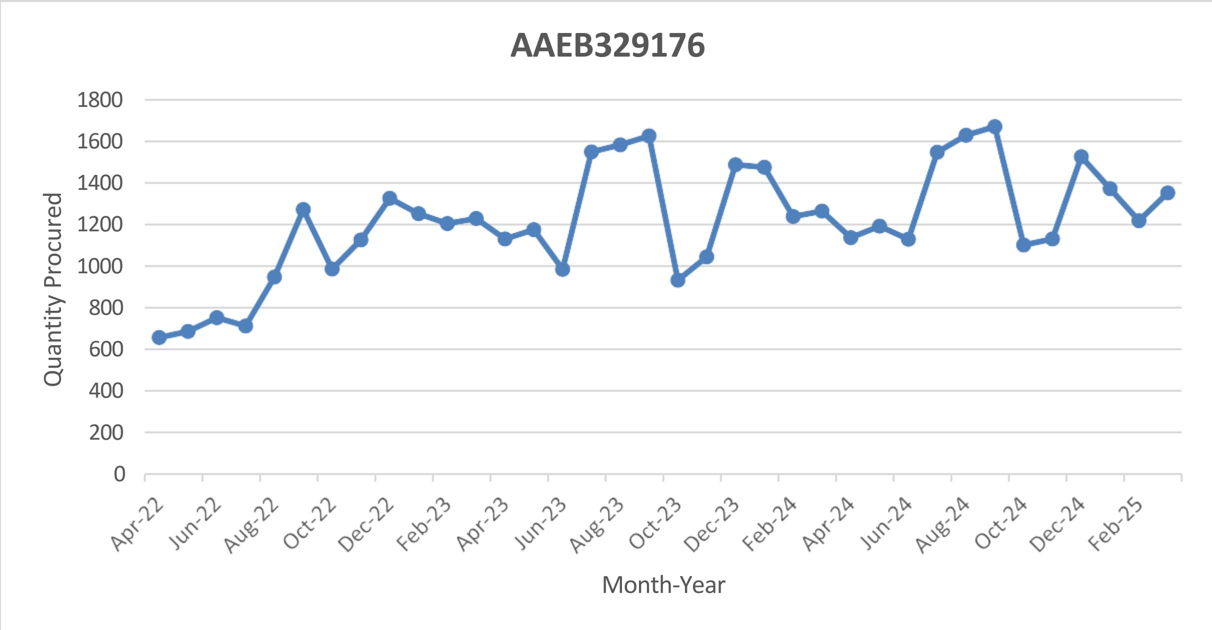
1. **Data Analysis**: The main aim of this step is to identify level, trend, seasonality and irregular behavior and also to identify if the time series is stationary or not. For this we'll plot data into different information charts to extract meaningful insights.

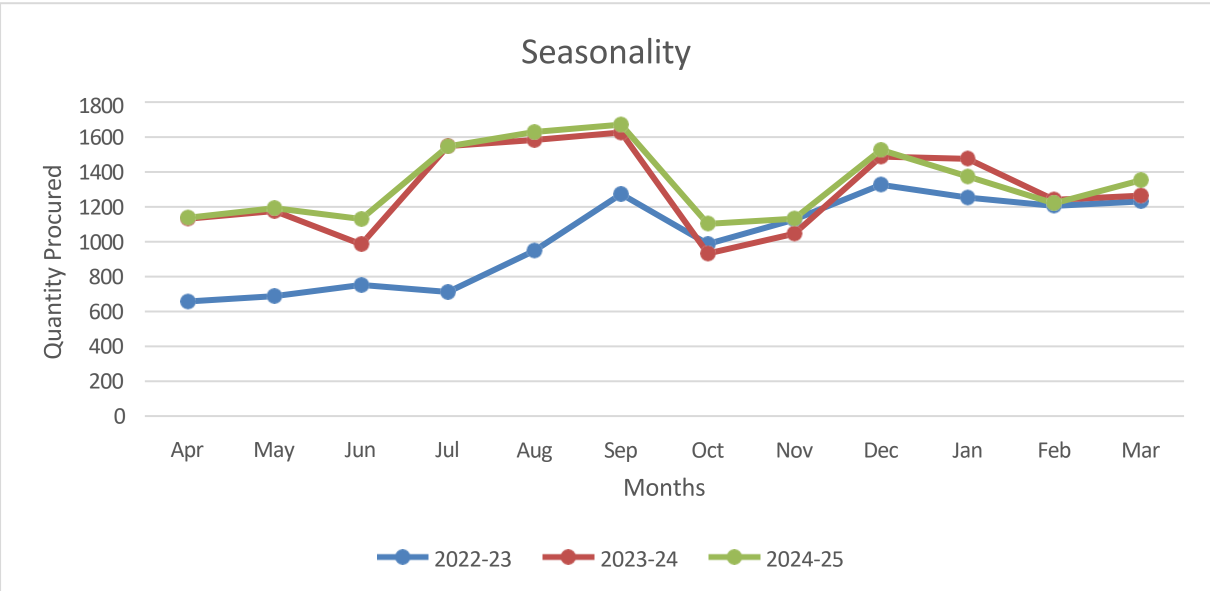
2. **Fitting Model**: Now we'll fit various statistical models on the data collected. The main aim of this step is to fit a model in order to forecast the future demand. To compare actual V/s predicted values we plot a line graph.

3. **Selecting Best Model**: Best model on the basis of 'MAPE' metric is used to compare the fitted models. Model having lowest MAPE is considered as best fitted model.

4. **Forecasting Demand**: The final step in process to forecast the demand for next period. We will forecast the demand data for upcoming year that will serve as a benchmark for the firm for their purchase orders next year.

i. **Data Analysis**





An initial examination of the Demand data reveals the presence of a **Trend component**, a **Seasonal component**, and some degree of **irregular variation**. This indicates that the demand series is **non-stationary**. As such, forecasting models capable of handling trend and seasonality are required.

To identify the most suitable forecasting method, we have selected two models for comparison:

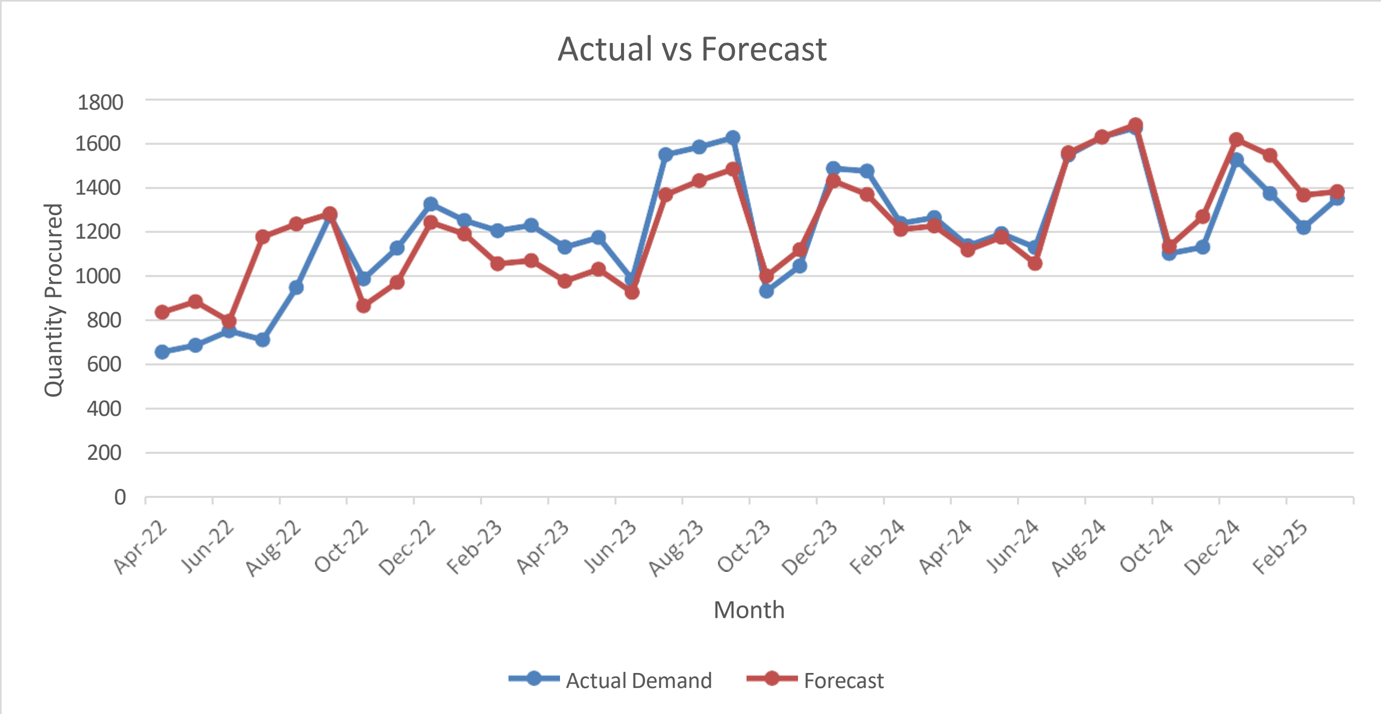
- **Decomposition Method**
- **Holt-Winters Exponential Smoothing**

Both models will be applied to the complete demand dataset. Their performance will be evaluated using the **Mean Absolute Percentage Error (MAPE)** metric. The model that yields the **lowest MAPE** will be considered the most accurate and will be used for final demand forecasting.

ii. Fitting Model

Model 1 : Decomposition Method

Time Period	Month-Year	Actual Demand	12 point Moving Average	Centered Moving Average	Ratio to Moving Average	Adjusted Seasonal Indices	Deseasonalize	Trend	Forecast	Error %
1	Apr-22	657				85.87	765.12	974.46	836.77	27.36
2	May-22	687				89.46	767.94	988.10	883.95	28.67
3	Jun-22	752				79.50	945.95	1001.73	796.34	5.90
4	Jul-22	712				116.02	613.71	1015.37	1177.98	65.45
5	Aug-22	948				120.05	789.69	1029.00	1235.29	30.30
6	Sep-22	1273	1012.92			122.96	1035.31	1042.63	1282.01	0.71
7	Oct-22	987	1052.42	1032.67	0.95	81.95	1204.36	1056.27	865.64	12.30
8	Nov-22	1126	1093.08	1072.75	1.04	90.78	1240.31	1069.90	971.29	13.74
9	Dec-22	1326	1112.50	1102.79	1.20	114.67	1156.40	1083.54	1242.45	6.30
10	Jan-23	1252	1182.25	1147.38	1.9	108.53	1153.59	1097.17	1190.76	4.89
11	Feb-23	1205	1235.17	1208.71	0.99	95.02	1268.09	1110.80	1055.54	12.40
12	Mar-23	1230	1264.58	1249.88	0.98	95.20	1292.07	1124.44	1070.42	12.97
13	Apr-23	1131	1260.00	1262.29	0.89	85.87	1317.12	1138.07	977.25	13.59
14	May-23	1175	1253.25	1256.63	0.93	89.46	1313.44	1151.71	1030.31	12.31
15	Jun-23	985	1266.67	1259.96	0.78	79.50	1239.05	1165.34	926.40	5.95
16	Jul-23	1549	1285.25	1275.96	1.21	116.02	1335.17	1178.97	1367.79	11.70
17	Aug-23	1583	1288.08	1286.67	1.23	120.05	1318.65	1192.61	1431.69	9.56
18	Sep-23	1626	1290.92	1289.50	1.26	122.96	1322.39	1206.24	1483.18	8.78
19	Oct-23	932	1291.42	1291.17	0.72	81.95	1137.25	1219.88	999.72	7.27
20	Nov-23	1045	1292.83	1292.13	0.80	90.78	1151.09	1233.51	1119.82	7.16
21	Dec-23	1487	1304.83	1298.83	1.14	114.67	1296.81	1247.14	1430.05	3.83
22	Jan-24	1475	1304.67	1304.75	1.13	108.53	1359.07	1260.78	1368.33	7.23
23	Feb-24	1239	1308.50	1306.58	0.94	95.02	1303.87	1274.41	1211.01	2.26
24	Mar-24	1264	1312.25	1310.38	0.96	95.20	1327.79	1288.05	1226.17	2.99
25	Apr-24	1137	1326.42	1319.33	0.86	85.87	1324.11	1301.68	1117.74	1.69
26	May-24	1192	1333.58	1330.00	0.89	89.46	1332.44	1315.31	1176.68	1.29
27	Jun-24	1129	1336.83	1335.21	0.84	79.50	1420.19	1328.95	1056.47	6.42
28	Jul-24	1547	1328.33	1332.58	1.16	116.02	1333.44	1342.58	1557.60	0.69
29	Aug-24	1629	1326.67	1327.50	1.22	120.05	1356.96	1356.22	1628.10	0.06
30	Sep-24	1671	1334.00	1330.33	1.25	122.96	1358.99	1369.85	1684.35	0.80
31	Oct-24	1102				81.95	1344.68	1383.48	1133.80	2.89
32	Nov-24	1131				90.78	1245.82	1397.12	1268.35	12.14
33	Dec-24	1526				114.67	1330.82	1410.75	1617.66	6.01
34	Jan-25	1373				108.53	1265.08	1424.39	1545.89	12.59
35	Feb-25	1219				95.02	1282.82	1438.02	1366.48	12.10
36	Mar-25	1352				95.20	1420.23	1451.65	1381.91	2.21



Model 2 : Holt Winter's Additive Method

Model Description

			Model Type
Model ID	VAR00001	Model_1	Winters' Additive

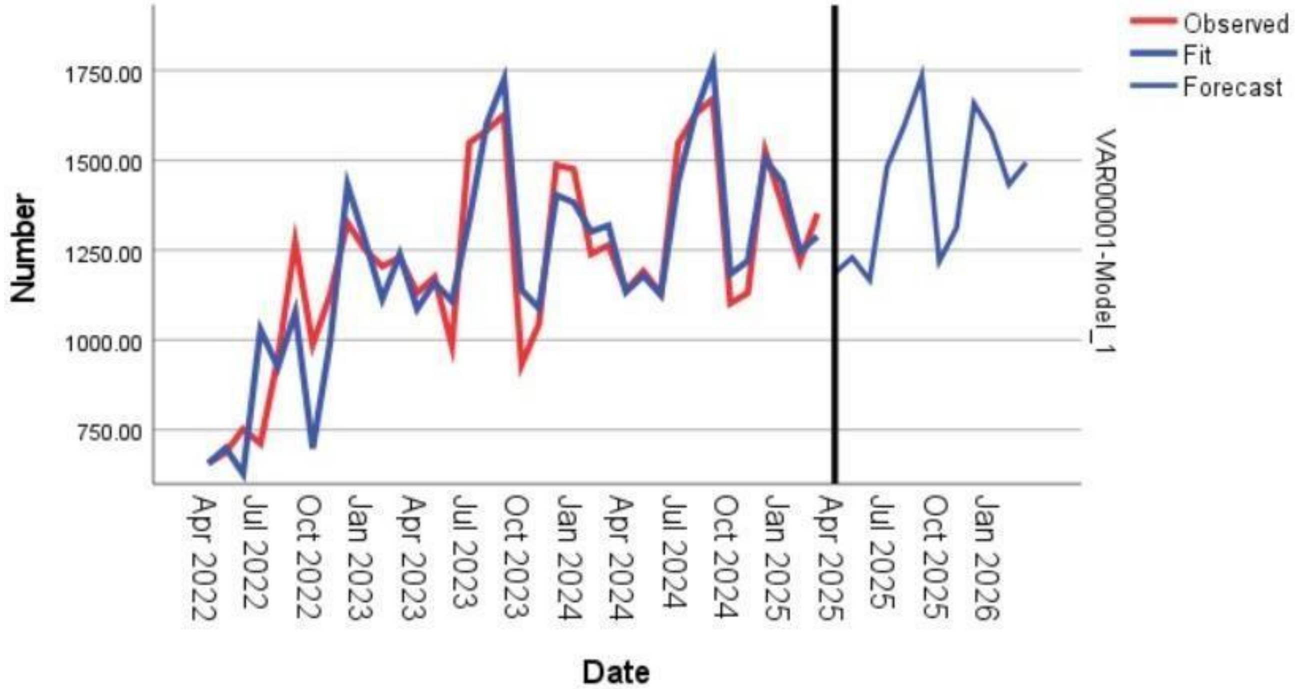
Model Summary

Model Fit

Fit Statistic	Mean	SE	Minimum	Maximum	Percentile 5	10	25	50	75	90	95
Stationary R-squared	.799	.	.799	.799	.799	.799	.799	.799	.799	.799	.799
R-squared	.823	.	.823	.823	.823	.823	.823	.823	.823	.823	.823
RMSE	118.213	.	118.213	118.213	118.213	118.213	118.213	118.213	118.213	118.213	118.213
MAPE	7.489	.	7.489	7.489	7.489	7.489	7.489	7.489	7.489	7.489	7.489
MaxAPE	44.511	.	44.511	44.511	44.511	44.511	44.511	44.511	44.511	44.511	44.511
MAE	82.234	.	82.234	82.234	82.234	82.234	82.234	82.234	82.234	82.234	82.234
MaxAE	316.921	.	316.921	316.921	316.921	316.921	316.921	316.921	316.921	316.921	316.921
Normalized BIC	9.844	.	9.844	9.844	9.844	9.844	9.844	9.844	9.844	9.844	9.844

Model Statistics

Model	Number of Predictors	Model Fit statistics: Stationary R-squared	MAPE	Ljung-Box Q(18): Statistics	DF	Sig.	Number of Outliers
VAR00001-Model_1	0	.799	7.489	22.352	15	.099	0



iii. **Selecting Best Model**

Method	MAPE
Decomposition Method	10.347%
Holt Winter's Additive Method	7.489%

As the Holt-Winters Additive Exponential Smoothing method produced the lowest Mean Absolute Percentage Error (MAPE) among the evaluated models, it has been selected as the most suitable approach for forecasting demand for the upcoming year.

iv. Forecasting Demand

Forecast

Model		Apr 2025	May 2025	Jun 2025	Jul 2025	Aug 2025	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mar 2026
VAR00001-Model_1	Forecast	1186.55	1229.55	1166.89	1480.90	1598.24	1734.91	1218.59	1312.26	1657.94	1578.28	1432.61	1493.61
	UCL	1427.06	1523.19	1505.43	1859.05	2012.24	2181.89	1696.29	1818.82	2191.81	2138.13	2017.30	2102.12
	LCL	946.05	935.91	828.35	1102.75	1184.24	1287.93	740.89	805.70	1124.07	1018.42	847.93	885.10

For each model, forecasts start after the last non-missing in the range of the requested estimation period, and end at the last period for which non-missing values of all the predictors are available or at the end date of the requested forecast period, whichever is earlier.

The forecasted demand values for the next year are:

Month	Forecast
Apr-25	1187
May-25	1230
Jun-25	1167
Jul-25	1481
Aug-25	1598
Sep-25	1735
Oct-25	1219
Nov-25	1312
Dec-25	1658
Jan-26	1578
Feb-26	1433
Mar-26	1494

1.3.3 Product 3:

ACEB405938

Using the sales data of past three years for product ACEB405938, we'll prepare a time series forecasting model based on the features of data.

Sales Data Table			
Month	2022-23	2023-24	2024-25
Apr	642	1146	1132
May	559	1159	1186
Jun	712	1330	1190
Jul	596	1340	1242
Aug	671	1355	1316
Sep	858	1378	1560
Oct	762	841	1020
Nov	852	866	1028
Dec	1066	1069	1123
Jan	1073	1142	1130
Feb	918	902	1078
Mar	1039	946	1112

Following is the stepwise approach that we will be following to solve our problem.

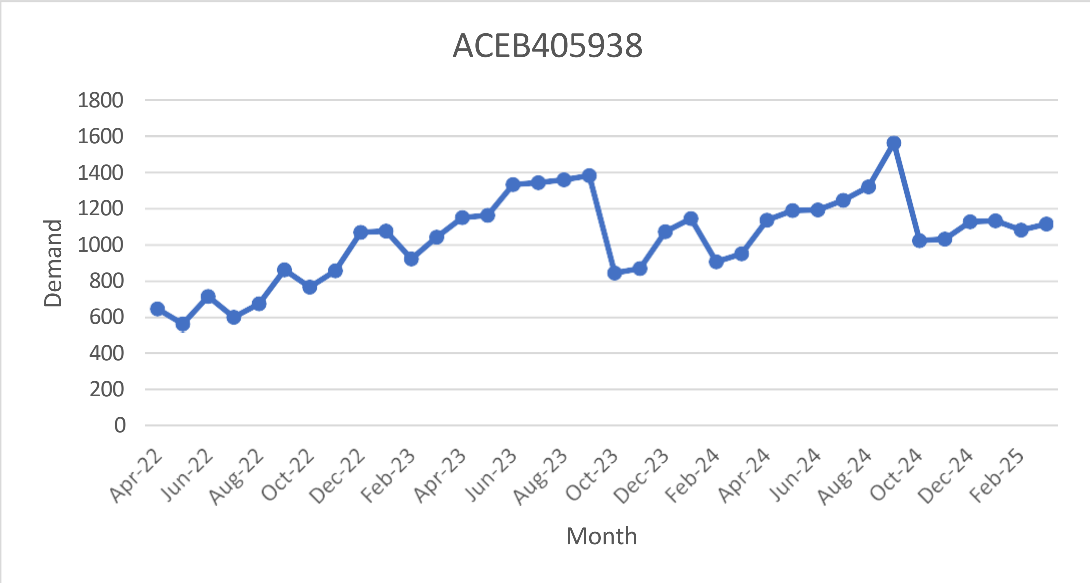
1. **Data Analysis**: The main aim of this step is to identify level, trend, seasonality and irregular behavior and also to identify if the time series is stationary or not. For this we'll plot data into different information charts to extract meaningful insights.

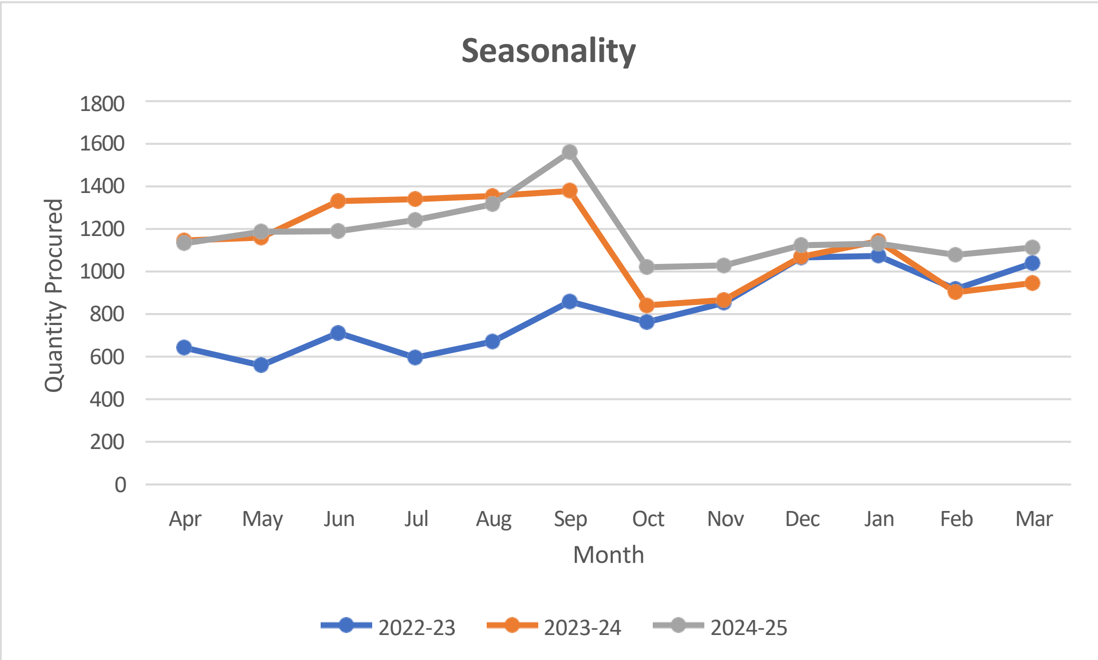
2. **Fitting Model**: Now we'll fit various statistical models on the data collected. The main aim of this step is to fit a model in order to forecast the future demand. To compare actual V/s predicted values we plot a line graph.

3. **Selecting Best Model**: Best model on the basis of 'MAPE' metric is used to compare the fitted models. Model having lowest MAPE is considered as best fitted model.

4. **Forecasting Demand**: The final step in process to forecast the demand for next period. We will forecast the demand data for upcoming year that will serve as a benchmark for the firm for their purchase orders next year.

i) Data Analysis





An initial examination of the Demand data reveals the presence of a **Trend component**, a **Seasonal component**, and some degree of **irregular variation**. This indicates that the demand series is **non-stationary**. As such, forecasting models capable of handling trend and seasonality are required.
To identify the most suitable forecasting method, we have selected two models for comparison:

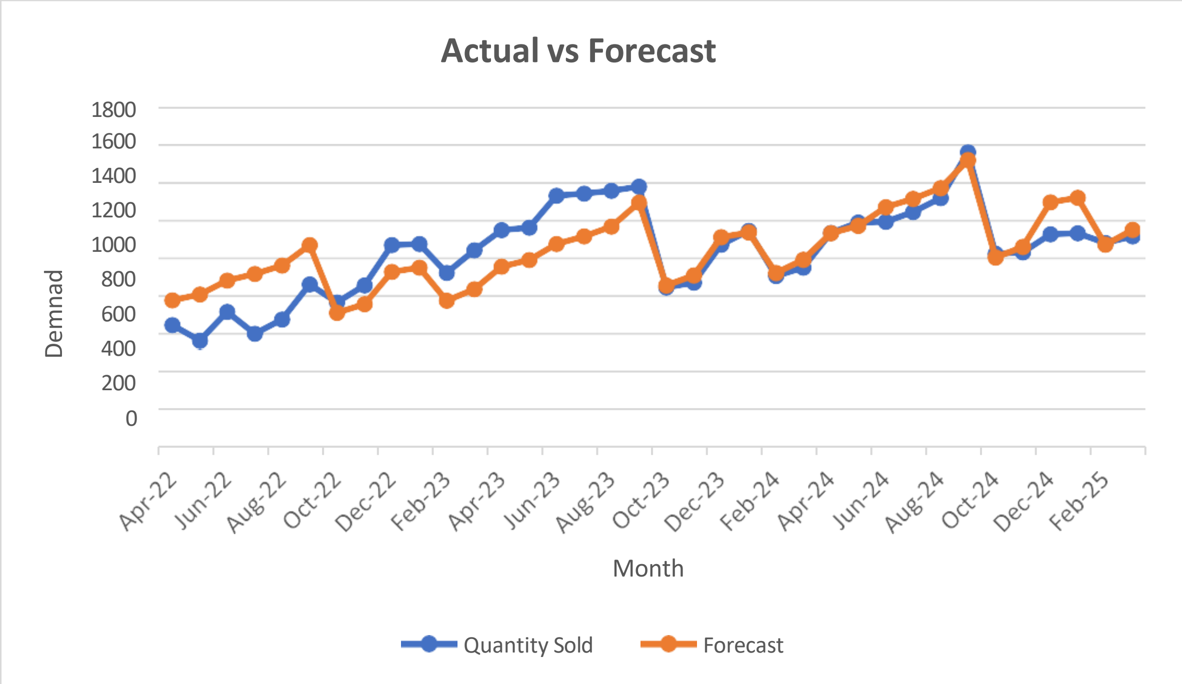
- **Decomposition Method**
- **Holt-Winters Exponential Smoothing**

Both models will be applied to the complete demand dataset. Their performance will be evaluated using the **Mean Absolute Percentage Error (MAPE)** metric. The model that yields the **lowest MAPE** will be considered the most accurate and will be used for final demand forecasting.

(ii) Fitting Model

Model 1 : Decomposition Method

Time Period	Month-Year	Quantity Sold	12 point Moving Average	Centered Moving Average	Ratio to Moving Average	Adjusted Seasonal Indices	Deseasonalize	Trend	Forecast	Error %
1	Apr-22	642				98.98	648.59	780.64	772.71	20.36
2	May-22	559				101.08	553.04	795.67	804.24	43.87
3	Jun-22	712				108.22	657.90	810.70	877.37	23.23
4	Jul-22	596				110.63	538.71	825.73	913.54	53.28
5	Aug-22	671				113.98	588.70	840.76	958.29	42.82
6	Sep-22	858	812.33			124.71	688.00	855.78	1067.25	24.39
7	Oct-22	762	854.33	833.33	8.33	81.15	939.02	870.81	706.65	7.26
8	Nov-22	852	904.33	879.33	8.79	84.88	1003.82	885.84	751.86	11.75
9	Dec-22	1066	955.83	930.08	9.30	102.53	1039.67	900.87	923.69	13.35
10	Jan-23	1073	1017.83	986.83	9.87	103.29	1038.78	915.90	946.07	11.83
11	Feb-23	918	1074.83	1046.33	10.46	82.70	1110.09	930.93	769.84	16.14
12	Mar-23	1039	1118.17	1096.50	10.97	87.84	1182.80	945.96	830.95	20.02
13	Apr-23	1146	1124.75	1121.46	11.21	98.98	1157.75	960.99	951.23	17.00
14	May-23	1159	1125.92	1125.33	11.25	101.08	1146.64	976.02	986.54	14.88
15	Jun-23	1330	1126.17	1126.04	11.26	108.22	1228.93	991.05	1072.55	19.36
16	Jul-23	1340	1131.92	1129.04	11.29	110.63	1211.20	1006.07	1113.06	16.94
17	Aug-23	1355	1130.58	1131.25	11.31	113.98	1188.80	1021.10	1163.85	14.11
18	Sep-23	1378	1122.83	1126.71	11.27	124.71	1104.96	1036.13	1292.16	6.23
19	Oct-23	841	1121.67	1122.25	11.22	81.15	1036.37	1051.16	853.00	1.43
20	Nov-23	866	1123.92	1122.79	11.23	84.88	1020.32	1066.19	904.94	4.50
21	Dec-23	1069	1112.25	1118.08	11.18	102.53	1042.59	1081.22	1108.60	3.70
22	Jan-24	1142	1104.08	1108.17	11.08	103.29	1105.58	1096.25	1132.36	0.84
23	Feb-24	902	1100.83	1102.46	11.02	82.70	1090.74	1111.28	918.98	1.88
24	Mar-24	946	1116.00	1108.42	11.08	87.84	1076.93	1126.31	989.37	4.58
25	Apr-24	1132	1130.92	1123.46	11.23	98.98	1143.61	1141.34	1129.75	0.20
26	May-24	1186	1144.42	1137.67	11.38	101.08	1173.35	1156.36	1168.83	1.45
27	Jun-24	1190	1148.92	1146.67	11.47	108.22	1099.57	1171.39	1267.73	6.53
28	Jul-24	1242	1147.92	1148.42	11.48	110.63	1122.62	1186.42	1312.59	5.68
29	Aug-24	1316	1162.58	1155.25	11.55	113.98	1154.59	1201.45	1369.41	4.06
30	Sep-24	1560	1176.42	1169.50	11.70	124.71	1250.90	1216.48	1517.07	2.75
31	Oct-24	1020				81.15	1256.96	1231.51	999.35	2.02
32	Nov-24	1028				84.88	1211.18	1246.54	1058.01	2.92
33	Dec-24	1123				102.53	1095.26	1261.57	1293.52	15.18
34	Jan-25	1130				103.29	1093.96	1276.60	1318.65	16.69
35	Feb-25	1078				82.70	1303.57	1291.63	1068.12	0.92
36	Mar-25	1112				87.84	1265.91	1306.65	1147.79	3.22



Model 2: Holt Winter's Additive Algorithm

Model Description

			Model Type
Model ID	VAR00001	Model_1	Winters' Additive

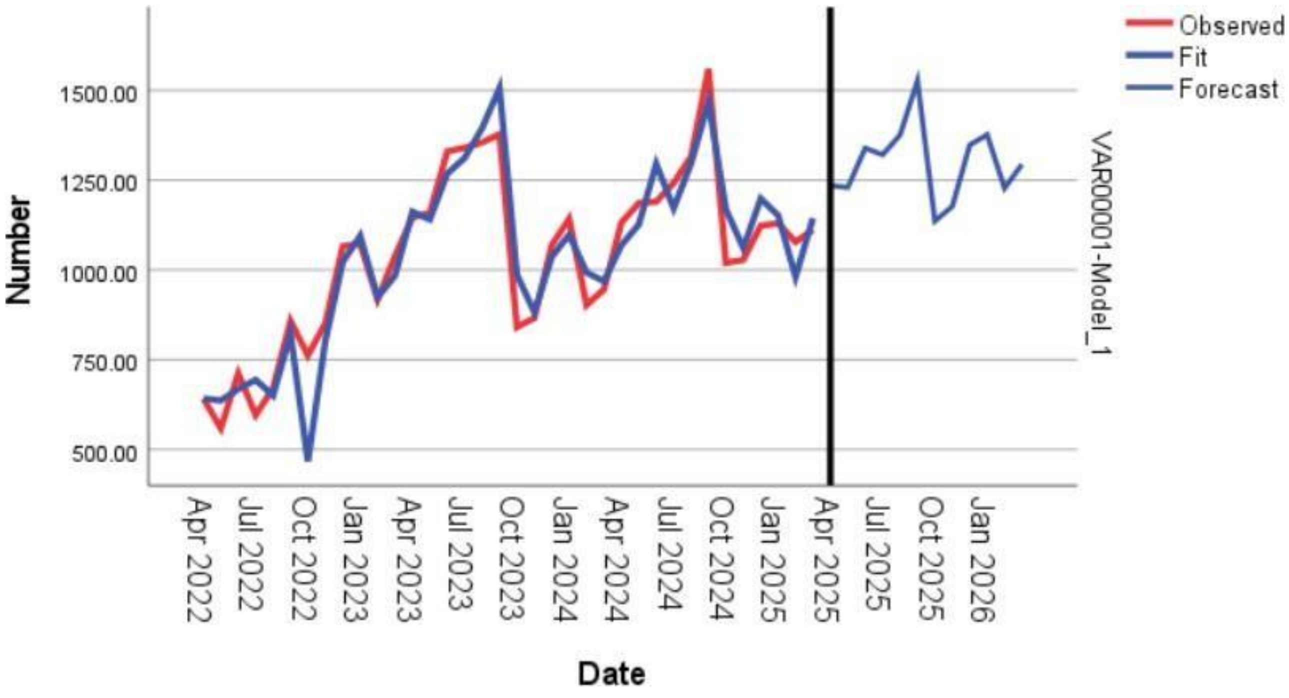
Model Summary

Model Fit

Fit Statistic	Mean	SE	Minimum	Maximum	Percentile 5	10	25	50	75	90	95
Stationary R-squared	.685	.	.685	.685	.685	.685	.685	.685	.685	.685	.685
R-squared	.877	.	.877	.877	.877	.877	.877	.877	.877	.877	.877
RMSE	85.815	.	85.815	85.815	85.815	85.815	85.815	85.815	85.815	85.815	85.815
MAPE	6.431	.	6.431	6.431	6.431	6.431	6.431	6.431	6.431	6.431	6.431
MaxAPE	38.716	.	38.716	38.716	38.716	38.716	38.716	38.716	38.716	38.716	38.716
MAE	61.328	.	61.328	61.328	61.328	61.328	61.328	61.328	61.328	61.328	61.328
MaxAE	295.017	.	295.017	295.017	295.017	295.017	295.017	295.017	295.017	295.017	295.017
Normalized BIC	9.203	.	9.203	9.203	9.203	9.203	9.203	9.203	9.203	9.203	9.203

Model Statistics

Model	Number of Predictors	Model Fit statistics: Stationary R-squared	MAPE	Ljung-Box Q(18): Statistics	DF	Sig.	Number of Outliers
VAR00001-Model_1	0	.685	6.431	19.246	15	.203	0



iii. Selecting Best Model

Method	MAPE
Decomposition Method	12.649%
Holt Winter's Additive Method	6.431%

As the Holt-Winters Additive Exponential Smoothing method produced the lowest Mean Absolute Percentage Error (MAPE) among the evaluated models, it has been selected as the most suitable approach for forecasting demand for the upcoming year.

iv. Forecasting Demand

Forecast

Model		Apr 2025	May 2025	Jun 2025	Jul 2025	Aug 2025	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mar 2026
VAR00001-Model_1	Forecast	1235.04	1229.71	1339.04	1321.04	1375.70	1527.04	1136.04	1177.03	1347.70	1376.70	1227.70	1294.03
	UCL	1409.64	1476.63	1641.47	1670.27	1766.18	1954.81	1598.11	1671.04	1871.70	1929.08	1807.07	1899.20
	LCL	1060.45	982.79	1036.61	971.80	985.23	1099.27	673.96	683.03	823.70	824.32	648.33	688.86

For each model, forecasts start after the last non-missing in the range of the requested estimation period, and end at the last period for which non-missing values of all the predictors are available or at the end date of the requested forecast period, whichever is earlier.

The forecasted demand values for the next year are:

Month	Forecast
Apr-25	1235
May-25	1230
Jun-25	1339
Jul-25	1321
Aug-25	1376
Sep-25	1527
Oct-25	1136
Nov-25	1177
Dec-25	1348
Jan-26	1377
Feb-26	1228
Mar-26	1294

1.3.4 Product 4:

ASP763504

Using the sales data of past three years for product ASP763504 **, we'll prepare a time series forecasting model based on the features of data.**

Sales Data Table			
Month	2022-23	2023-24	2024-25
Apr	612	1112	1084
May	589	1120	1091
Jun	621	1157	1136
Jul	786	1506	1555
Aug	741	1598	1625
Sep	972	1667	1724
Oct	893	963	1007
Nov	1196	1228	1168
Dec	955	1043	1037
Jan	979	1074	1061
Feb	1225	1329	1416
Mar	1234	1462	1529

Following is the stepwise approach that we will be following to solve our problem.

1. **Data Analysis**: The main aim of this step is to identify level, trend, seasonality and irregular behavior and also to identify if the time series is stationary or not. For this we'll plot data into different information charts to extract meaningful insights.

2. **Fitting Model**: Now we'll fit various statistical models on the data collected. The main aim of this step is to fit a model in order to forecast the future demand. To compare actual V/s predicted values we plot a line graph.

3. **Selecting Best Model**: Best model on the basis of 'MAPE' metric is used to compare the fitted models. Model having lowest MAPE is considered as best fitted model.

4. **Forecasting Demand**: The final step in process to forecast the demand for next period. We will forecast the demand data for upcoming year that will serve as a benchmark for the firm for their purchase orders next year.